# afterVideoLesson3.2.pdf

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Paralelismo

3º Grado en Ingeniería Informática

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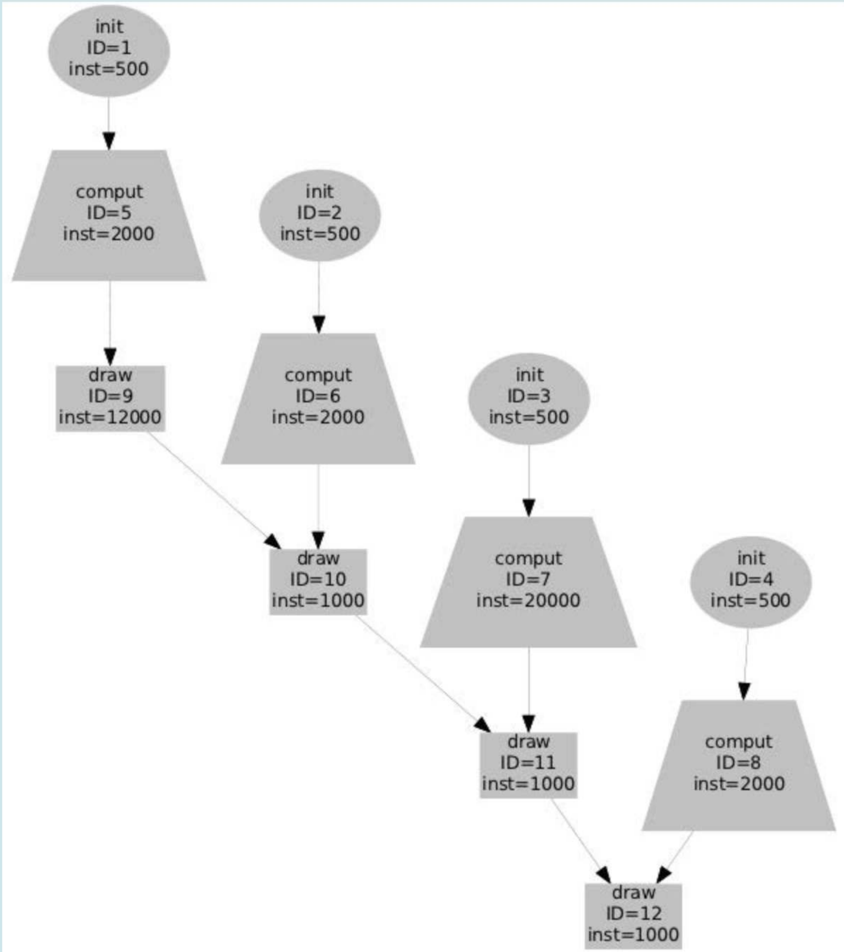
| | |
|---|---|
| **Qualificació** | **4,00** sobre 4,00 (**100**%) |

**Pregunta 1**
Completa
Puntuació 1,00 sobre 1,00
Marca la pregunta

Given the following task dependence graph (TDG). Each node is labeled with a name and also includes an identifier *ID* and cost *inst* in terms of the number of instructions.




The TDG can also be expressed textually with the following table.


The TDG can also be expressed textually with the following table:

| name | ID | inst | succesor ID task |
|---|---|---|---|
| init | 1 | 500 | 5 |
| init | 2 | 500 | 6 |
| init | 3 | 500 | 7 |
| init | 4 | 500 | 8 |
| compute | 5 | 2000 | 9 |
| compute | 6 | 2000 | 10 |
| compute | 7 | 20000 | 11 |
| compute | 8 | 2000 | 12 |
| draw | 9 | 12000 | 10 |
| draw | 10 | 1000 | 11 |
| draw | 11 | 1000 | 12 |
| draw | 12 | 1000 | No successor |

Observe that task *comput* with *ID=7* takes 10 times more that the other *comput* tasks and task *draw* with *ID=9* takes 12 times more than other *draw* tasks.

Assume that the tasks in the TDG are executed on 4 processors with the following task assignment: each processor executes a sequence *init-compute-draw* (for example the sequence {2, 6, 10}). Which is the speed-up that is obtained?

Trieu-ne una:

- 1.91
- 2.45
- 4

Pregunta 2
Completa
Puntuació 1,00 sobre 1,00
Marca la pregunta

Assuming that we are able to better balance the work among processors, which means that each node 1-4 weights 500, each node 5-8 weights 6500, and each node 9-12 weights 3750. Which is the speed-up that would be achieved with 4 processors, assuming the same task assignment as before?

Trieu-ne una:

- 4
- 1.95
- 4.77

Pregunta 3
Completa
Puntuació 1,00 sobre 1,00
Marca la pregunta

Assume a sequential application computing the sum of two vectors of size N=1024 elements. Which should be the problem size and task granularity when parallelized with P=4 processors and strong scaling:

Trieu-ne una:

- 1024 and 256, respectively.
- 1024 and 1024, respectively.
- 4096 and 1024, respectively.

**Pregunta 4**

Completa

Puntuació 1,00 sobre 1,00

Marca la pregunta

Which should be the problem size and task granularity when parallelized with P=4 processors and weak scaling:

Trieu-ne una:

- ○ 1024 and 256, respectively.
- ○ 4096 and 256, respectively.
- ◉ 4096 and 1024, respectively.